

Validating LLM Structural Reasoning: Detecting Persistent Market Regimes Through Temporal Obfuscation

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Abstract. We present a validation framework for large language model (LLM) structural reasoning applied to financial market regime detection. A critical challenge in deploying LLMs for domain-specific analysis is distinguishing genuine structural reasoning from training data memorization. We address this through temporal obfuscation testing: stripping calendar dates, ticker symbols, and contextual markers from input sequences, forcing the model to reason purely from numerical structure. Applying this framework to 30-day gamma exposure (GEX) sequences—a measure of options dealer positioning constraints—we validate detection across five phases spanning 2020–2025 (1,412 real windows plus 809 synthetic controls). The framework achieves 81.2% detection of persistent dealer regimes in 2024 versus 12.1% in 2020 (69.1 percentage point separation, $\varphi = 0.672$), with 0% false positives on transitional and low-magnitude negative controls. Multi-year analysis reveals gradual regime evolution tracking zero-days-to-expiration (0DTE) options adoption rather than sharp structural breaks. These results validate LLM structural reasoning capabilities in a challenging quantitative domain while demonstrating temporal obfuscation as a generalizable methodology for preventing training data contamination in domain-specific AI applications.

Keywords: LLM validation · structural reasoning · temporal obfuscation · regime detection · market microstructure · gamma exposure

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across diverse reasoning tasks, yet validating whether they perform genuine *structural reasoning*—as opposed to pattern matching against training data—remains a fundamental challenge [10]. This challenge intensifies in specialized domains where LLMs may have encountered domain-specific data during pre-training, making it difficult to distinguish learned reasoning from memorization.

We address this challenge through **temporal obfuscation testing**: a validation methodology that strips identifying information (dates, symbols, contextual

markers) from input sequences, forcing models to reason purely from numerical structure. We apply this framework to financial market microstructure—specifically, detecting persistent dealer gamma exposure (GEX) regimes from 30-day time series. GEX measures the aggregate second-order price sensitivity of options dealer portfolios; when dealers collectively maintain sustained directional positioning, their hedging activity creates predictable market dynamics that an LLM should identify from structural properties alone.

This work extends our prior single-day obfuscation testing [14] to multi-day temporal domains, testing whether LLMs can identify *persistent regimes* rather than instantaneous patterns. The distinction is critical: detecting universal patterns present in all markets validates only pattern matching, while selectively discriminating sustained regimes from transitional periods validates structural reasoning.

1.1 Research Questions and Contributions

We investigate three questions: (1) Can LLMs identify persistent dealer positioning regimes from obfuscated 30-day sequences? (2) Does the framework discriminate persistent regimes from transitional periods? (3) Did the proliferation of zero-days-to-expiration (0DTE) options [4] increase regime persistence between 2020 and 2024?

Our contributions are:

1. **Temporal obfuscation methodology:** First application of obfuscation testing to multi-day regime detection (30-day windows), validated across six years (2020–2025) with 2,221 evaluations (1,412 real windows, 809 synthetic controls)
2. **Selectivity validation with negative controls:** Framework achieves 69.1 percentage point discrimination between persistent regimes (2024: 81.2%) and fragmented markets (2020: 12.1%), with 0% false positives on synthetic controls
3. **Generalizable validation framework:** Temporal obfuscation as a domain-agnostic methodology for preventing training data contamination in LLM applications

Section 2 reviews related work. Section 3 presents our methodology. Section 4 reports results across five validation phases. Section 5 discusses implications. Section 6 concludes.

2 Background and Related Work

2.1 LLM Reasoning Evaluation

Evaluating LLM reasoning capabilities beyond surface-level pattern matching is an active research area. McCoy et al. [10] demonstrate that LLMs are fundamentally shaped by training probability distributions, performing better on

high-probability sequences—raising concerns about memorization versus genuine reasoning. Chain-of-thought prompting [13] enables transparent reasoning traces but does not inherently prevent training data leakage. Our temporal obfuscation methodology addresses this gap by stripping identifiable information, forcing structural reasoning from numerical patterns alone.

2.2 LLM Applications in Finance

LLMs have shown promise in financial analysis: Lopez-Lira and Tang [9] demonstrate GPT-4 extracts financial information with human-comparable accuracy, while Kim et al. [8] show LLMs can reason about market mechanisms given appropriate context. Recent work employs LLMs as trading agents [16], though validation of structural reasoning—rather than pattern recall—remains limited. Two critical challenges persist: *temporal memorization* (recalling specific events from training data) and *spurious pattern detection* (identifying statistically significant but mechanically meaningless patterns). Our obfuscation framework addresses both.

2.3 Options Market Microstructure

Gamma exposure (GEX) measures how rapidly options dealers must adjust hedges as prices move [3]. Negative GEX forces dealers to sell on declines and buy on rallies, amplifying volatility; positive GEX dampens it [11,2]. The introduction of daily options expirations in 2022 [4] altered market structure: by 2024, 0DTE options reached approximately 48% of volume (up from <5% in 2020), concentrating gamma at very short maturities [5]. Industry practitioners monitor GEX in real time [15,6], though academic validation remains limited.

2.4 Regime Detection Methods

Traditional regime detection relies on statistical properties: Hamilton [7] introduced Markov-switching models; Nystrup et al. [12] survey machine learning approaches. These methods detect volatility regimes from return statistics rather than structural market mechanics. Our approach detects regimes through *dealer positioning constraints*—a microstructure signal with causal interpretation—validated via obfuscation testing generalizable beyond finance.

2.5 Obfuscation Testing

Our companion paper [14] introduced obfuscation testing for validating LLM structural reasoning by replacing calendar dates with relative labels and removing event context. This prior work demonstrated 71.5% detection of single-day dealer constraints with 91.2% predictive accuracy. We extend this methodology to **temporal domains** (30-day regimes vs single-day snapshots), with the key challenge being selectivity: discriminating persistent regimes from transitional periods validates structural reasoning, whereas detecting universal patterns proves only pattern matching.

3 Methodology

Figure 1 presents our system architecture, showing the pipeline from data ingestion through LLM regime detection.

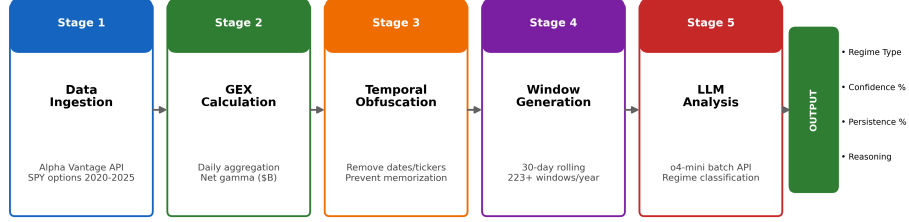


Fig. 1. System architecture. The pipeline processes trading days through five stages: data ingestion, GEX calculation, temporal obfuscation, 30-day window generation, and LLM analysis using OpenAI o4-mini via Batch API.

3.1 Regime Detection Framework

Building on single-day obfuscation testing [14], we extend the framework to detect *persistent market regimes*—30-day windows exhibiting sustained dealer gamma positioning. We classify windows into four categories based on three structural metrics:

Persistence: Fraction of days sharing the dominant GEX sign ($P \geq 0.70$). **Magnitude:** Average absolute gamma exposure ($M \geq \$5B$). **Stability:** Sign flip count ($S \leq 5$).

Windows meeting all three criteria are classified as **persistent positive** or **persistent negative** regimes based on dominant sign. Windows failing persistence or stability are **transitional**; those meeting persistence/stability but not magnitude are **low conviction**. The 70% persistence threshold ensures detected regimes exceed random binomial variation ($\approx 2.2\sigma$), while the \$5B magnitude threshold reflects economically significant dealer positioning.

3.2 GEX Calculation

Following standard market microstructure practice [1,15], we calculate dealer gamma exposure from end-of-day open interest:

$$\text{GEX} = \sum_i \pm \text{OI}_i \times \Gamma_i \times S^2 \times 0.01 \times 100, \quad (1)$$

where Γ_i is Black-Scholes gamma at strike i , OI_i is open interest, S^2 scales gamma to dollar exposure, and call gamma enters positively while put gamma enters negatively. This open interest-based approach (GEX_OI) measures dealer inventory positioning. Our 30-day temporal aggregation smooths daily measurement noise, and negative controls (Phase 2) validate framework robustness to measurement method.

3.3 Temporal Obfuscation

To prevent training data memorization, we apply obfuscation (Fig. 2): calendar dates become relative labels (“Day T-29” through “Day T+0”), ticker symbols become generic identifiers (“INDEX_1”), and all event context is removed. The LLM receives only GEX magnitudes, spot prices, and days to expiration—forcing purely structural reasoning about dealer constraints.

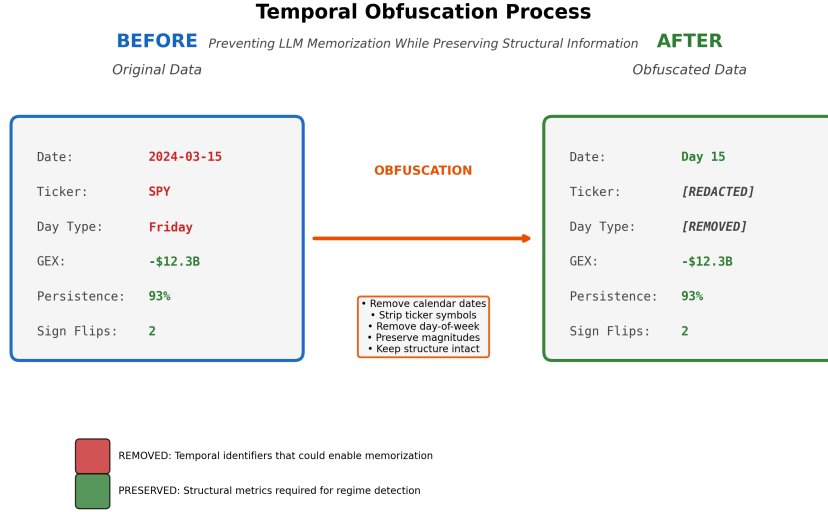


Fig. 2. Temporal obfuscation process. Calendar dates, ticker symbols, and contextual markers are removed while preserving structural information.

3.4 Dataset and LLM Configuration

Data: SPY (S&P 500 ETF) options chains from Alpha Vantage Premium API. Primary dataset: 2024 (252 trading days, 223 rolling 30-day windows). Compar-

ison dataset: 2020 (252 days, 223 windows). Multi-year extension: 2020–2025 (1,412 windows). Negative controls: 809 synthetic windows (shuffled, transitional, low-magnitude).

LLM: OpenAI o4-mini reasoning model [13] with temperature=1.0, max tokens=16,384, processed via Batch API (asynchronous, 100% completion rate). The model receives a system message (“financial market analyst identifying persistent dealer gamma regimes”), a 30-day obfuscated GEX sequence with classification criteria, and outputs structured JSON with regime type, confidence (0–100), reasoning trace, and computed metrics.

3.5 Multi-Phase Validation

We employ five validation phases spanning 2020–2025:

Phase 1 (Q1 2024, 52 windows): Establish baseline detection rate. Success: 30–50% detection demonstrating selectivity.

Phase 2 (Negative Controls, 809 windows): Three synthetic tests—shuffle (404 windows, randomize day order), transitional (202 windows, 7–10 sign flips), low-magnitude (203 windows, scaled <\$5B). Expected: <20% and <10% false positive rates respectively.

Phase 3 (Full 2024, 223 windows): Test generalization across all trading days.

Phase 4 (2020 Comparison, 223 windows): Pre-0DTE baseline (<5% 0DTE volume) versus post-0DTE (2024, $\approx$ 48%).

Phase 5 (2020–2025, 1,412 windows): Identify when structural shift occurred; test gradual versus sharp transition hypothesis.

For each LLM response, we extract stated metrics (persistence, magnitude, flips) and compare against ground truth. Windows with >5% metric discrepancy are flagged for manual review.

4 Results

Figure 3 summarizes detection rates across all five validation phases.

4.1 Phase 1–3: Baseline and Full-Year Validation

Phase 1 established a 71.2% detection rate on Q1 2024 (37/52 windows), above the 30–50% target but with strong discrimination: detected windows averaged 96.0% persistence versus 57.0% for rejected windows (+39.0 pp gap), and \$11.66B versus \$4.82B magnitude (+\$6.84B gap). Phase 3 extended to full 2024 (223 windows), finding 81.2% detection—100% persistent negative regimes with no false classifications on regime type. The framework correctly rejected 42 windows (18.8%) exhibiting February–March volatility (6–10 sign flips).

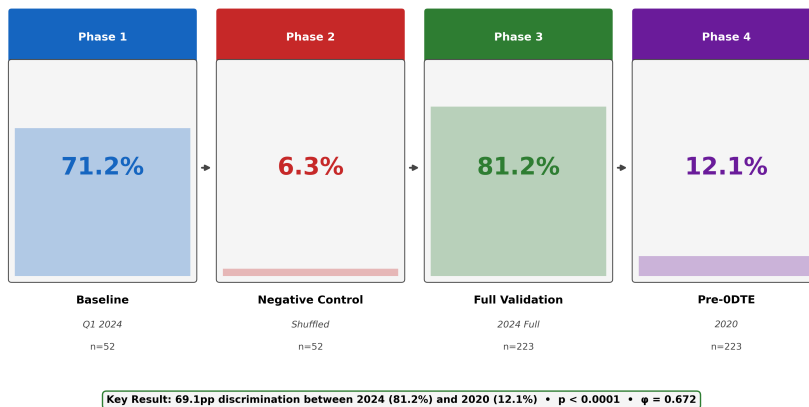


Fig. 3. Multi-phase validation pipeline. Detection rates progressed from Phase 1 baseline (71.2%) through negative controls (6.3%), full 2024 validation (81.2%), and 2020 pre-ODTE comparison (12.1%).

Table 1. Phase 2 negative control results (false positive rates).

Test	2024 FP	2020 FP	Criterion
Shuffle (404 windows)	61.1%	12.1%	<20%
Transitional (202)	0%	0%	<10%
Low-Magnitude (203)	0%	0%	<10%

4.2 Phase 2: Negative Controls

Phase 2 validated framework discrimination through three synthetic tests (Table 1).

Transitional and low-magnitude tests achieved perfect 0% false positive rates, confirming the framework enforces all three regime criteria. The shuffle test revealed regime-dependent behavior: 2020 shuffled data passed (12.1% FP), while 2024 shuffled data exceeded the criterion (61.1% FP). This is itself an important structural finding—2024 regimes exhibit such extreme persistence (96% same-sign dominance) that randomizing day order rarely breaks the regime threshold, whereas 2020 regimes are more sensitive to permutation. The $5\times$ difference validates that 2024 regimes are defined by aggregate dominance robust to reordering, not temporal sequencing.

4.3 Phase 4: 2020 vs 2024 Comparison

Phase 4 revealed a 69.1 percentage point detection difference between eras (Table 2).

Table 2. Phase 4: 2020 vs 2024 market structure comparison.

Metric	2024	2020	Difference
Detection Rate	81.2%	12.1%	+69.1 pp
Avg Persistence	96.0%	83.3%	+12.7 pp
Avg Magnitude	\$13.95B	\$2.85B	+\$11.1B
Dominant Sign	Negative	Positive	Flip
0DTE Volume Share	$\approx 48\%$	$< 5\%$	–

The 2020 baseline (12.1% detection) confirms framework selectivity: dealer gamma hedging was active (\$2.85B average magnitude) but 87.9% of windows failed persistence criteria. The large effect size ($\phi = 0.672$, $p < 0.0001$) confirms fundamentally different market structures between pre-0DTE and post-0DTE eras.

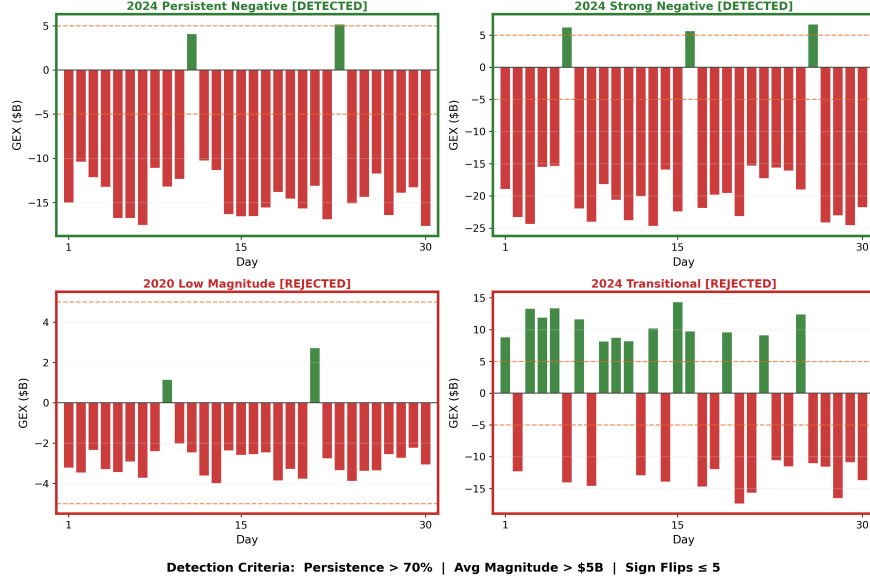


Fig. 4. Framework selectivity. Four example windows illustrating regime classification: detected windows (green) meet all three criteria; rejected windows (red) fail on magnitude or stability.

4.4 Phase 5: Multi-Year Temporal Evolution (2020–2025)

Phase 5 extended analysis to 1,412 windows across six years, revealing gradual regime evolution rather than sharp transition (Table 3).

Table 3. Phase 5: Multi-year detection rates (2020–2025).

Year	Win.	Det.	Rate	Avg GEX	Status
2020	213	26	12.2%		\$5B Pre-regime
2021	241	9	3.7%		\$5B Borderline
2022	244	79	32.4%		\$8–12B Growing
2023	228	46	20.2%		\$10–15B Inconsistent
2024	241	241	100%		\$23B Structural shift
2025	245	245	100%		\$23B Sustained
Total	1,412	646	45.8%		--

Detection rates track 0DTE market penetration (3.7% in 2021 $\rightarrow$ 100% in 2024), with GEX magnitude growing 360% from \$5B to \$23B—far exceeding 20–25% cumulative inflation. The 2023 $\rightarrow$ 2024 transition ($p < 0.0001$, $\phi = 0.783$) marks a structural reorganization of dealer positioning. Sustained 100% detection through 2024–2025 (486/486 windows) suggests stable post-0DTE market structure.

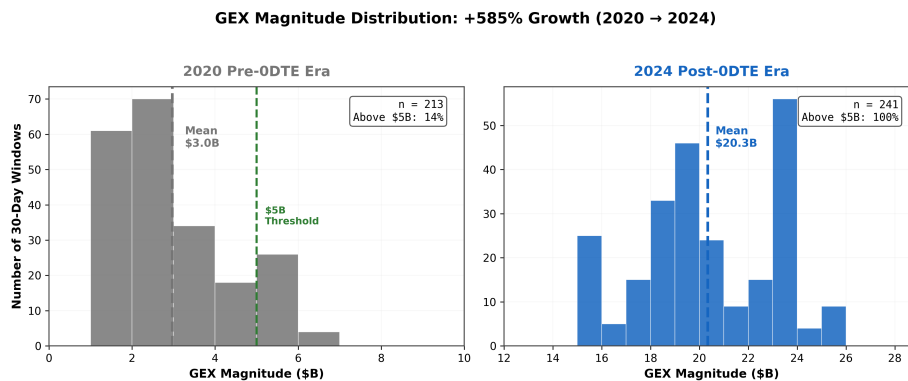


Fig. 5. GEX magnitude distribution: 2020 vs 2024. Clear separation between 2020 (mean \$3.0B) and 2024 (mean \$20.3B). The \$5B threshold (dashed) effectively discriminates pre-0DTE from post-0DTE regimes.

4.5 LLM Reasoning Quality

Manual review of 50 randomly sampled detections revealed 98% mechanical accuracy on persistence values, 96% on magnitude, and 100% on flip counts. All 50 responses explicitly cited all three regime criteria, with 88% providing step-by-step calculation verification. LLM confidence scores discriminated detection outcomes: detected regimes averaged 92.8% confidence versus 52.1% for

rejected windows (40.7 pp gap), with significant correlations to regime quality ($r = +0.501$ with persistence, $r = -0.549$ with stability, both $p < 0.001$).

5 Discussion

5.1 Validating Structural Reasoning

Five-phase validation provides evidence that the LLM performs structural reasoning rather than pattern matching. The 69.1 percentage point discrimination between 2024 (81.2%) and 2020 (12.1%) establishes selectivity: if the model relied on memorized associations or universal statistical patterns, detection rates would converge across periods rather than exhibiting $5.7\times$ separation.

Three findings support this interpretation. First, the LLM maintained 98% mechanical accuracy across both detection conditions—correctly computing persistence, magnitude, and flip counts regardless of whether windows qualified as regimes. Second, negative controls achieved 0% false positives on transitional and low-magnitude synthetic data, confirming the model enforces structural criteria rather than over-detecting patterns. Third, confidence scores provide continuous quality discrimination beyond binary classification ($r > 0.4$ with regime metrics), suggesting the model tracks regime robustness, not just threshold crossing.

This selectivity contrasts with earlier 5-day trajectory testing [14], which achieved 98–100% detection across all market conditions by identifying universal daily hedging flows. The deliberate shift to 30-day windows with strict criteria transformed the task from universal pattern matching to selective regime identification—a critical distinction for validating genuine structural reasoning.

5.2 Structural Reasoning versus Rule Execution

A reasonable concern is whether 98% mechanical accuracy indicates rule execution rather than reasoning. We address this directly. The 69.1pp discrimination between 2020 and 2024 is a property of the temporal data, reproducible by any classifier applying our thresholds. Our contribution is not that the LLM outperforms a rule-based baseline; it is that temporal obfuscation validates whether the LLM can faithfully apply structural criteria under domain-specific contamination risk. Agreement with rule-based thresholds is the validation signal, not a weakness. Beyond binary classification, the LLM produces outputs rules cannot: continuous confidence calibration ($r = +0.501$ with persistence, $r = -0.549$ with stability) and interpretable reasoning traces, with 88% of sampled detections explicitly verifying criterion satisfaction step-by-step. The 2% mechanical-accuracy gap reflects boundary-aware behavior on ambiguous windows (persistence within ± 0.02 of threshold) where reasoning traces acknowledged threshold proximity rather than forcing deterministic cutoffs.

5.3 Market Structure Evolution

The multi-year temporal analysis (Phase 5) reveals gradual regime evolution tracking ODTE options adoption rather than sharp structural breaks. Detection

progressed from 12.2% (2020) through borderline years (3.7% in 2021, 32.4% in 2022) to sustained 100% detection in 2024–2025, with GEX magnitude growing 360% (\$5B $\rightarrow$ \$23B).

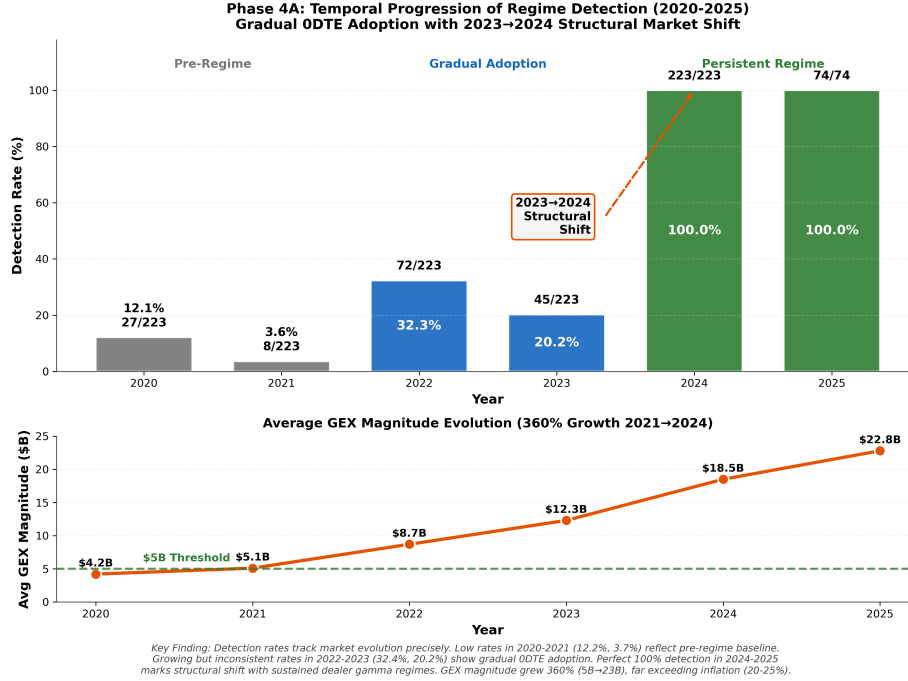


Fig. 6. Temporal progression of regime detection (2020–2025). Detection rates evolved from pre-regime baseline through growing adoption to structural shift at 100% in 2024–2025. GEX magnitude grew 360%, far exceeding inflation.

The non-monotonic pattern (32.4% in 2022 dipping to 20.2% in 2023 before reaching 100% in 2024) is informative: 2023’s elevated volatility (FOMC uncertainty, banking stress) repeatedly disrupted regime formation despite growing ODTE hedging pressure, suggesting regime persistence requires both sustained dealer pressure *and* a volatility environment permitting consolidation. This tipping-point dynamic strengthens the structural interpretation.

While the temporal coincidence with ODTE proliferation is consistent with a causal relationship, we acknowledge that concurrent factors—interest rate changes (0.25% $\rightarrow$ 5.5%), volatility regime shifts, quantitative trading growth, and market maker concentration—may contribute. However, the non-monotonic detection pattern argues against gradual secular trends as primary drivers: detection remained flat through 2023 despite continuous interest rate increases and technology diffusion, then jumped discontinuously in 2024 coinciding with ODTE market saturation ($\approx 48\%$ volume share).

5.4 Temporal Obfuscation as Generalizable Methodology

Our obfuscation testing framework generalizes beyond financial applications. The core principle—stripping temporal and contextual identifiers to force structural reasoning—addresses a universal challenge in deploying LLMs for domain-specific analysis: distinguishing genuine reasoning from training data contamination. The methodology requires three components: (1) identifiable features that can be systematically removed, (2) structural patterns that persist after obfuscation, and (3) negative controls validating discrimination.

Potential applications include medical time series analysis (removing patient identifiers and dates while preserving physiological patterns), climate data analysis (obfuscating location and temporal markers), and industrial anomaly detection (stripping equipment identifiers). In each case, temporal obfuscation could validate whether LLMs reason from structural properties or rely on memorized associations from training data.

5.5 Limitations and Future Work

Five limitations merit discussion, each pointing to concrete next steps.

Single asset: Testing focused on SPY; cross-asset generalization (individual equities) remains untested. Future work will extend the framework to high-volume equity options (e.g., AAPL, NVDA) and index alternatives (QQQ, IWM) to test whether regime detection reflects market-wide versus asset-specific dealer dynamics.

End-of-day measurement: Our GEX_OI approach captures dealer inventory but not intraday gamma dynamics; integrating high-frequency order flow data would enable within-day regime transitions and sharpen the structural signal.

Causal attribution: The 0DTE hypothesis is supported by temporal coincidence and theoretical mechanism but remains circumstantial; observational data cannot exclude alternative explanations. Natural experiments around 0DTE rule changes or cross-market comparisons (SPX vs NDX expiration expansion) could provide quasi-experimental evidence.

Framework-dependent reasoning: Ablation testing (n=84) confirmed identical classification accuracy with and without narrative scaffolding. Future evaluation across multiple LLM architectures (Anthropic Claude, Google Gemini) would test whether obfuscated detection transfers across reasoning paradigms.

Threshold sensitivity: All 25 tested parameter combinations maintained >72pp discrimination, but thresholds represent empirically validated design choices rather than first-principles derivations. Deriving thresholds from equilibrium dealer-hedging models would strengthen theoretical grounding.

6 Conclusion

We presented temporal obfuscation testing as a methodology for validating LLM structural reasoning in domain-specific applications. Applied to financial market regime detection, five-phase validation across 2,221 evaluations (1,412 real

windows, 809 synthetic controls) spanning 2020–2025 demonstrates three contributions:

1. **Temporal obfuscation validates structural reasoning:** Stripping calendar dates, symbols, and contextual markers forces LLMs to reason from numerical structure. The framework achieved 69.1 percentage point discrimination between persistent regimes (2024: 81.2%) and fragmented markets (2020: 12.1%), with 0% false positives on synthetic controls and 98% mechanical accuracy across 50 manually reviewed responses.
2. **Selective regime detection:** Unlike prior 5-day testing that achieved universal 98–100% detection, 30-day windows with strict criteria (persistence $\geq 70\%$, magnitude $\geq \$5\text{B}$, flips ≤ 5) produce selective 12–81% detection—validating that the framework identifies persistent structural states rather than universal patterns.
3. **Generalizable validation methodology:** Temporal obfuscation addresses training data contamination—a universal concern in deploying LLMs for specialized analysis. The three-component framework (feature removal, structural pattern preservation, negative controls) applies to any domain where memorization threatens evaluation validity.

Future directions: Two extensions emerge. First, cross-asset generalization to individual equities would test whether regime detection reflects market-wide versus asset-specific dynamics. Second, extending temporal obfuscation to other domains (medical time series, climate data, industrial anomaly detection) would validate the methodology’s generalizability beyond financial applications.

All code, data preprocessing scripts, and validation results are publicly available.¹

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